

ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-I (Session: 2025-26)

Roll No.:

Course Name: B. Tech First Year

Subject Name: Design Thinking, Innovation & Entrepreneurship

Subject Code: 25HU102

Max. Marks: 15

Time: 75 Minutes

Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions		(1 x 5)		
(a)	What is the core idea of Design Thinking Concept?	CO1	K2	4.1.1
(b)	List 5 Stages of Design Thinking.	CO1	K2	4.1.1
(c)	List two essential skill required for Design Thinkers.	CO1	K2	4.1.1
(d)	Why is user feedback critical in testing phase of Design thinking?	CO2	K2	4.1.1
(e)	Define 'user-centric' approach in context of design thinking Process.	CO2	K2	2.2.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question		(Same K Levels) (2 x 2)		
(a)	Mention advantages of multi-disciplinary teamwork in Design Thinking with suitable example or scenario.	CO1	K2	3.2.1
(b)	Write at least 4 business uses of Design Thinking with examples and possible outcomes.	CO1	K2	2.2.1
(c)	List three physical tools and two digital tools that can be used in Design Thinking workshops.	CO1	K2	2.2.1
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Apply 5 stage model of design thinking to Jeevan Bindi.	CO1	K3	2.1.1
(b)	Apply 5 stage model of design thinking to Tata Nano.	CO1	K3	3.1.3
4. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Apply 5 stage model of design thinking to Embrace Infant warmer.	CO2	K3	3.2.3
(b)	Apply 5 stage model of design thinking to Lucky Iron Fish.	CO2	K3	3.2.3

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

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ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-II (Session: 2025-26)

Roll No.:

2 [REDACTED]

Course Name: B.Tech

Subject Name: Design Thinking, Innovation and Entrepreneurship

Subject Code: 25HU102

Max. Marks: 15

Time: 75 Minutes

Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions (1 x 5)				
(a)	Explain <i>shadowing</i> in empathy research.	CO3	K2	3.1.1
(b)	Mention the three essential components of "HMW" questions.	CO3	K2	3.1.1
(c)	What was the main aim of launching Big Basket?	CO3	K2	3.2.1
(d)	At which stage of Design Thinking does testing occur?	CO2	K2	3.1.1
(e)	What is low-fidelity prototyping?	CO2	K2	3.2.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question (Same K Levels) (2 x 2)				
(a)	Explain the characteristics of effective Design Thinking tools and their contribution to innovation.	CO3	K3	3.2.3
(b)	Explain the SCAMPER method and consider how it contributes to creative problem-solving.	CO3	K3	3.2.3
(c)	Discuss how low-fidelity and high-fidelity prototyping complement each other in the Design Thinking process.	CO3	K3	3.2.3
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Describe how Royal Enfield became a worldwide lifestyle brand from a failing motorbike business.	CO3	K4	5.1.1
(b)	Discuss how IDEO applied the principles of human-centered design (HCD) in developing the water pump.	CO3	K4	5.1.1
4. Attempt Any one Question (Same K Levels) (3 x 1)				
(a)	Apply five stage model of Design Thinking in the case study of GE VScan ultrasound device.	CO2	K3	3.2.2
(b)	Differentiate convergent and divergent thinking approaches and illustrate their application in design thinking process.	CO2	K3	3.2.3

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

**ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)**

Pre - End Term Exam (Session: 2025-26)

Admission No. [REDACTED]

Course Name: B.Tech.

Subject Name: Design Thinking, Innovation and Entrepreneurship

Subject Code: 25HU102

Max. Marks: 20

Time: 90 Minutes

Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Knowledge Level: K2 & K3)				
Q. No.	Question	CO	KL	PI
Attempt any one part from each question (3 Marks x 4 Questions = 12 Marks)				
1.(a)	Enumerate the differences between invention and innovation.	CO4	K3	3.2.1
1.(b)	List down the benefits of innovation.	CO4	K3	3.2.1
2.(a)	What is organizational innovation? Give suitable example.	CO4	K3	3.2.1
2.(b)	Differentiate between product and process innovation with suitable examples.	CO4	K3	3.2.1
3.(a)	Define entrepreneurship and list out its key characteristics.	CO5	K3	3.2.1
3.(b)	Explain copyright and the types of works protected under it.	CO5	K3	3.2.1
4.(a)	Differentiate between GI tag and trademark with suitable examples.	CO5	K3	3.2.1
4.(b)	List out common myths and realities associated with entrepreneurship. (any five)	CO5	K3	3.2.1
SECTION-B (Knowledge Level: K3)				
Attempt any one part from each question (4 Marks x 2 Questions = 8 Marks)				
5.(a)	What is the importance of TRL? Explain with a suitable example and diagram.	CO4	K3	3.2.2
5.(b)	Differentiate between social and grassroots innovation with suitable examples.	CO4	K3	3.2.2
6.(a)	Define pitch deck and explain its importance. Enumerate the key components of an effective pitch deck.	CO5	K3	3.2.2
6.(b)	Design a Business Model Canvas for a start-up idea of your choice and explain each component. Also, develop a strong value proposition addressing specific customer problem.	CO5	K3	3.2.2

CO Course Outcomes mapped with respective question

KL Bloom's Knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create